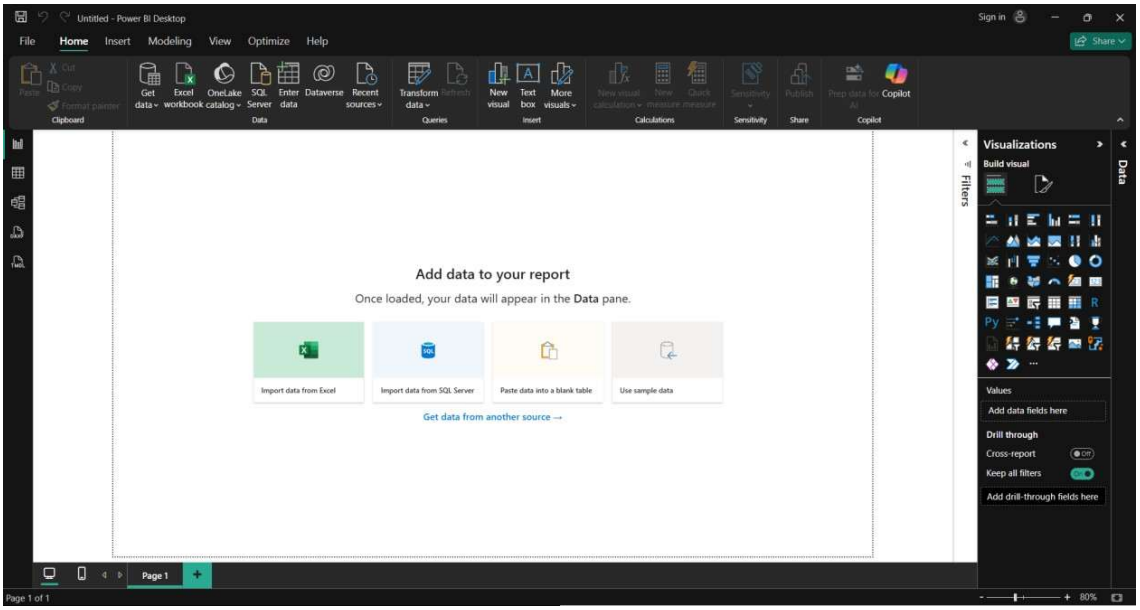
DR. D. Y. PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH Akurdi, Pune-411044 Department-Artificial Intelligence and Data Science		
Laboratory Journal | Subject: Computer Laboratory – IV		**AY:2025-26**
Name: Asad Alwase	**Class:** B.E. (A)	**Roll No:** 22542

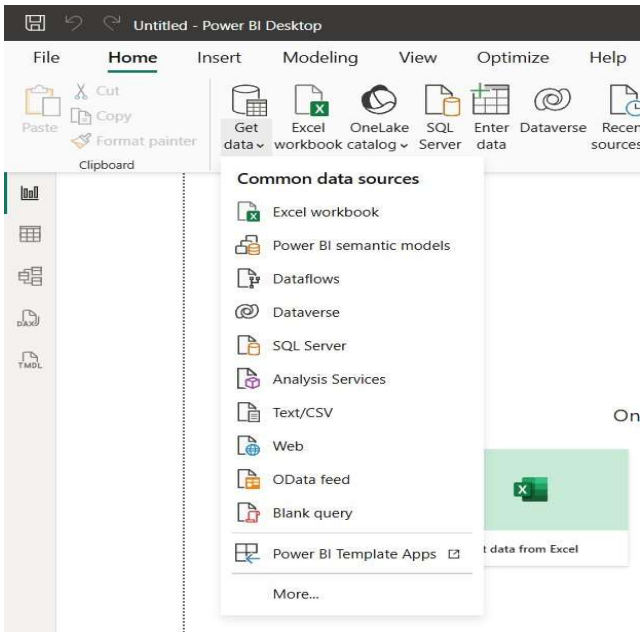
Lab Practical 7

Import Data from different Sources such as (Excel, Sql Server, Oracle etc.) and load in targeted system.

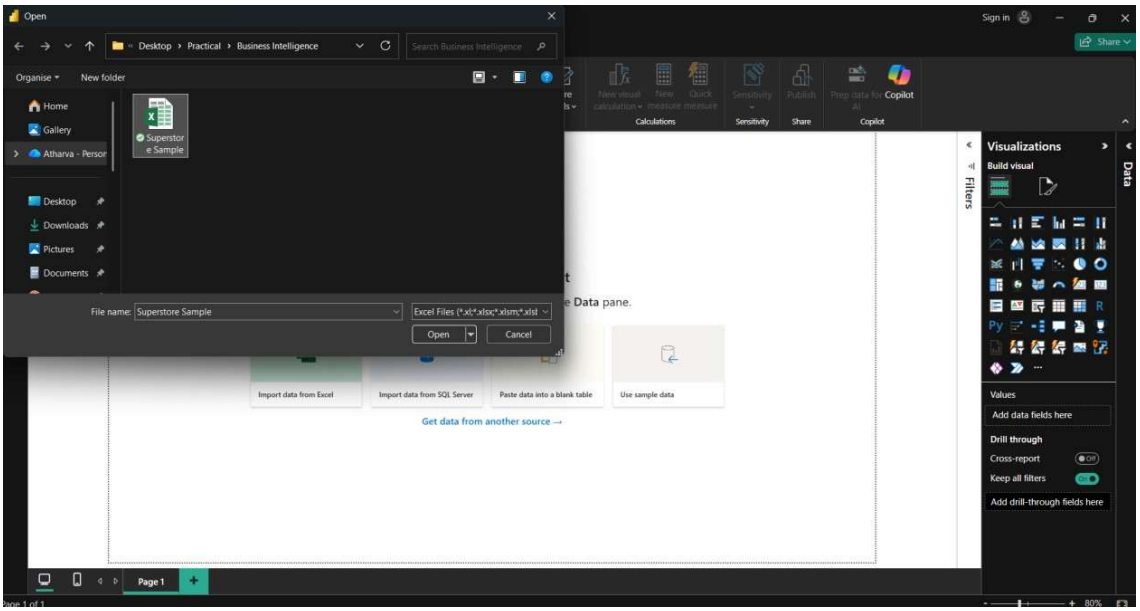
1) Launch Power BI Desktop.



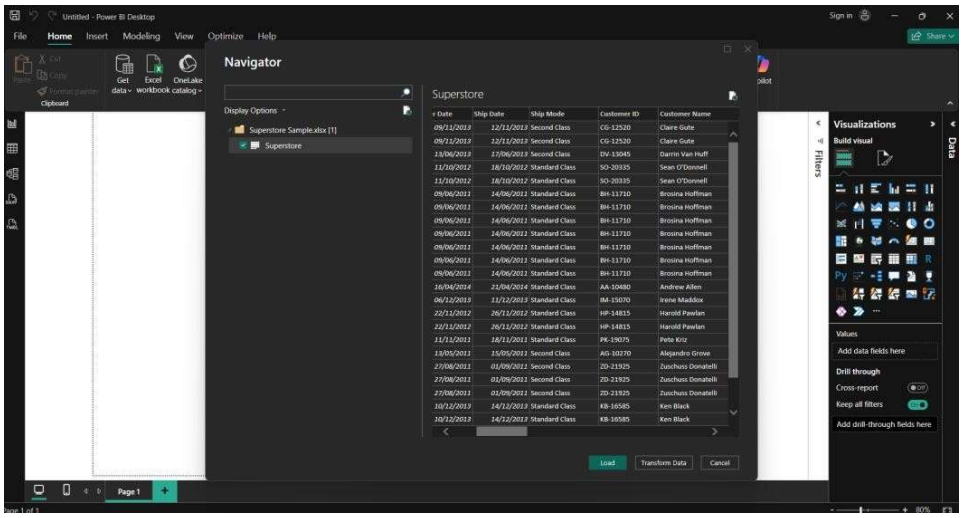
2) From the Home ribbon, select Get Data.

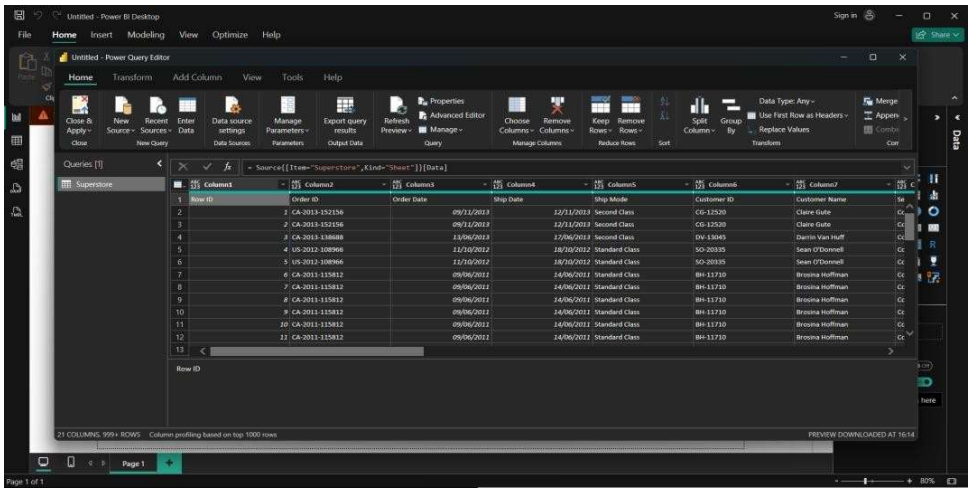


3) In the Open File dialog box, select the Products.xlsx file.



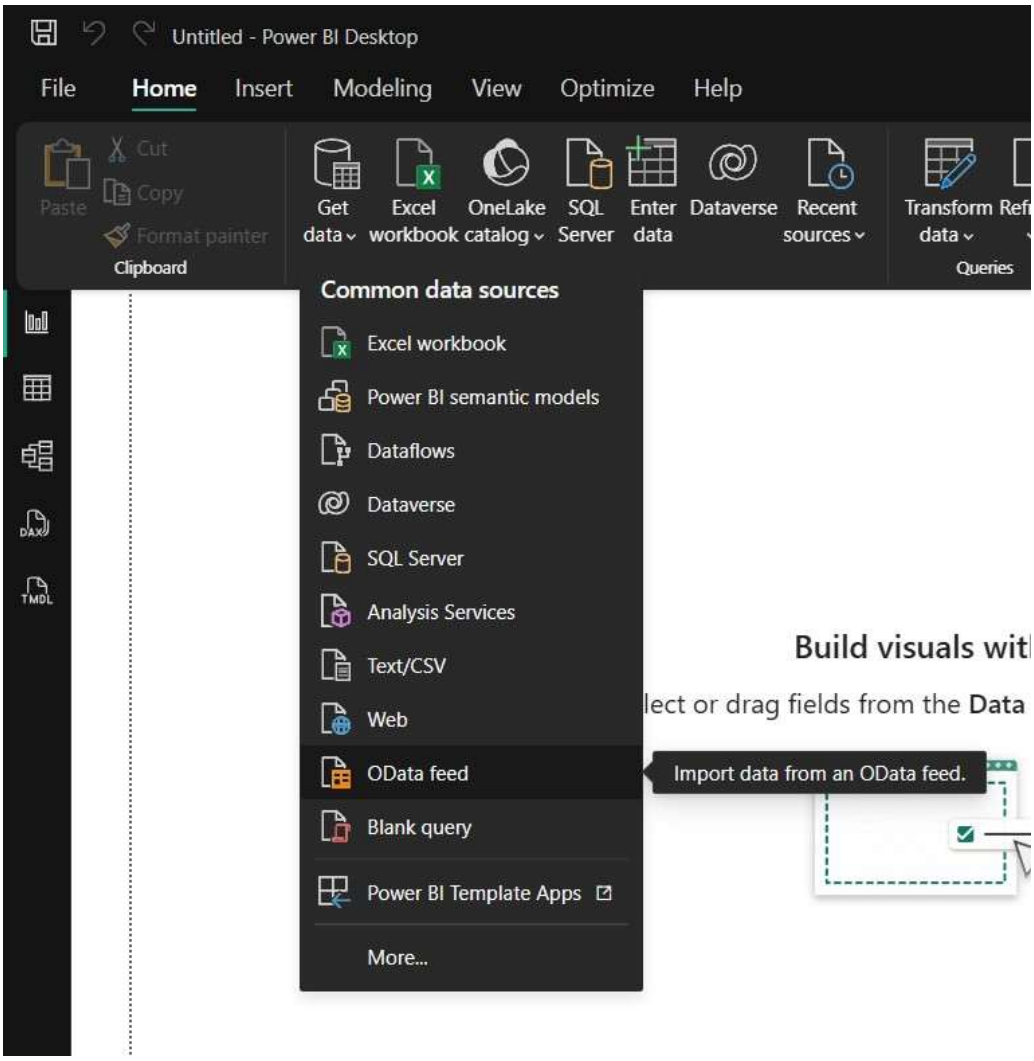
4) In the Navigator pane, select the Products table and then select Edit.



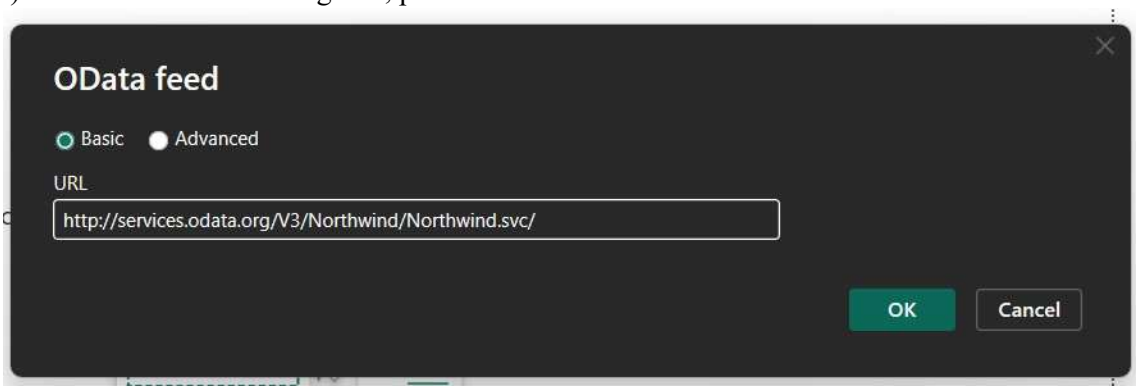


Importing Data from OData Feed
(http://services.odata.org/V3/Northwind/Northwind.svc/)

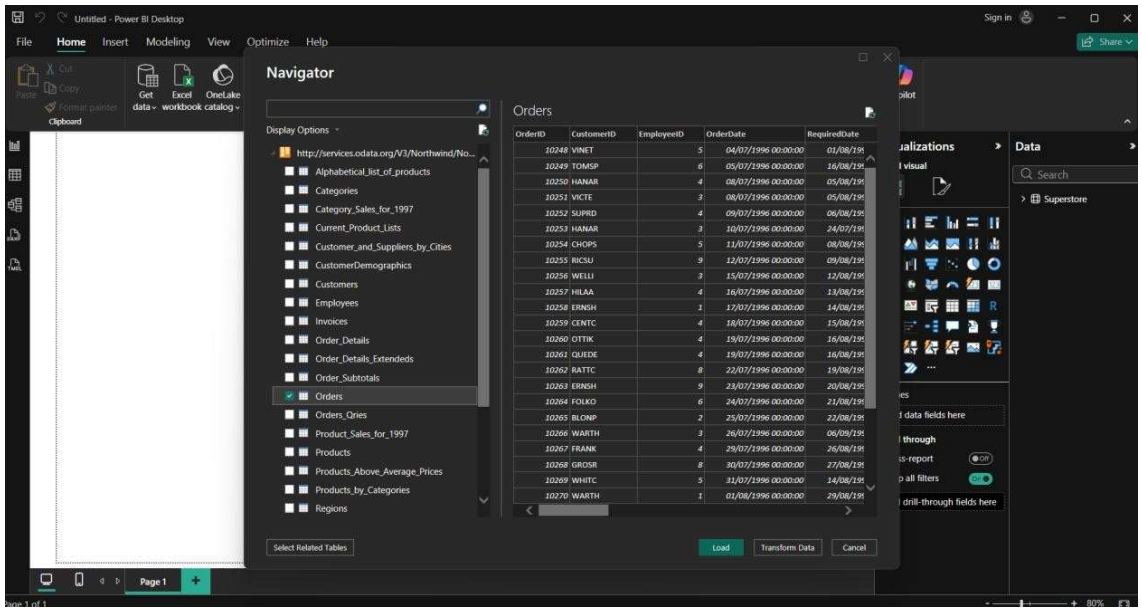
1) From the Home ribbon tab in Query Editor, select Get Data. Browse to the OData Feed data source.

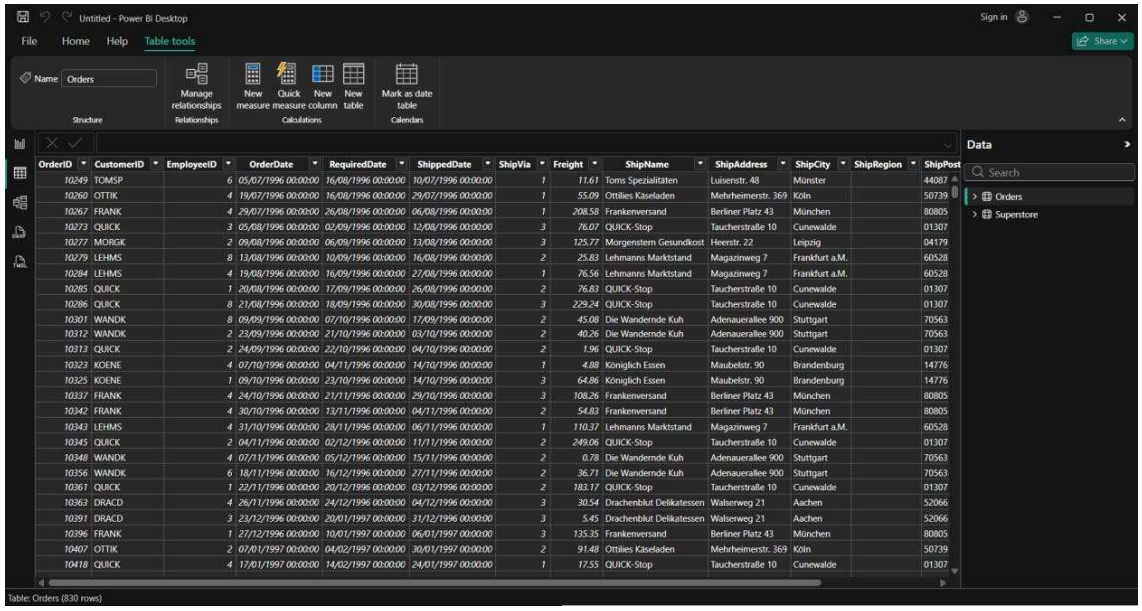


2) In the OData Feed dialog box, paste the URL for the Northwind OData feed.



3) In the Navigator pane, select the Orders table, and then select Edit


Untitled - Power BI Desktop
File
Home
Insert
Modeling
View
Optimize
Help
Navigator
Display Options
Orders
Select Related Tables
Load
Transform Data
Cancel
Page 1
Data
Superstore


Untitled - Power BI Desktop
File
Home
Help
Table tools
Name
Orders
Manage relationships
New measure
Quick measure
New column
New table
Mark as date table
Structure
Relationships
Calculations
Calendars
Data
Orders
Superstore
Table: Orders (830 rows)

Sign: _____.

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Lab Practical 8

Data Visualization from Extraction Transformation and Loading (ETL) Process.

Step 1: Check Data (Extraction)

```
import pandas as pd
import matplotlib.pyplot as plt

# EXTRACT - Load raw data from CSV
df = pd.read_csv("Superstore.csv", encoding='latin1')

print("Shape of data:", df.shape)
print("\nFirst 5 rows:")
df.head()
```

✓ 5.2s

```
Shape of data: (9994, 21)
```

Step 2: Transform Check-missing values

```
# Check missing values
print("Missing values:\n", df.isnull().sum())

# Drop rows with null values
df = df.dropna()

# Select only required columns
df = df[['Category', 'Sub-Category', 'Sales', 'Profit', 'Quantity']]
```

✓ 0.0s

```
Missing values:
 Row ID           0
Order ID         0
Order Date       0
Ship Date        0
Ship Mode        0
Customer ID      0
Customer Name    0
Segment          0
Country          0
City             0
State            0
Postal Code      0
Region           0
Product ID       0
Category         0
Sub-Category     0
Product Name     0
Sales            0
Quantity         0
Discount         0
Profit           0
dtype: int64
```

Remove Duplicates if any

```python
df = df.drop_duplicates()

print("\nCleaned Data Shape:", df.shape)
print(df.head())
```

✓ 0.3s

```
Cleaned Data Shape: (7729, 5)
         Category Sub-Category     Sales    Profit  Quantity
0       Furniture    Bookcases  261.9600   41.9136         2
1       Furniture       Chairs  731.9400  219.5820         3
2  Office Supplies       Labels   14.6200    6.8714         2
3       Furniture       Tables  957.5775 -383.0310         5
4  Office Supplies      Storage   22.3680    2.5164         2
```

Step 3: Load

```python
# LOAD - Save cleaned data to new CSV
df.to_csv("Superstore_cleaned.csv", index=False)

print("Cleaned data loaded/saved successfully!")
```

✓ 0.0s

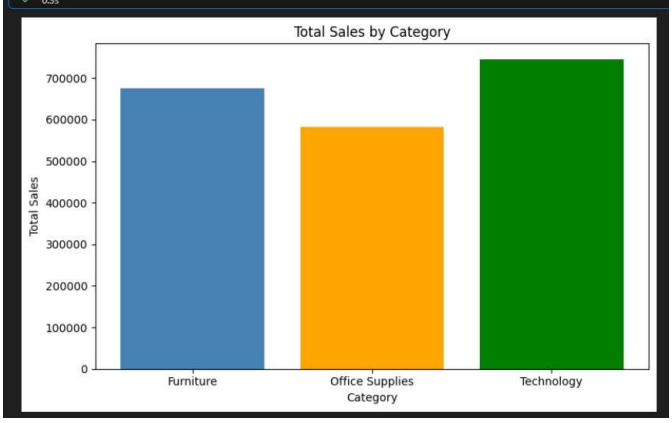
```
Cleaned data loaded/saved successfully!
```

Step 4: Visualization

```python
category_sales = df.groupby('Category')['Sales'].sum()

plt.figure(figsize=(8,5))
plt.bar(category_sales.index, category_sales.values, color=['steelblue','orange','green'])
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.tight_layout()
plt.show()
```

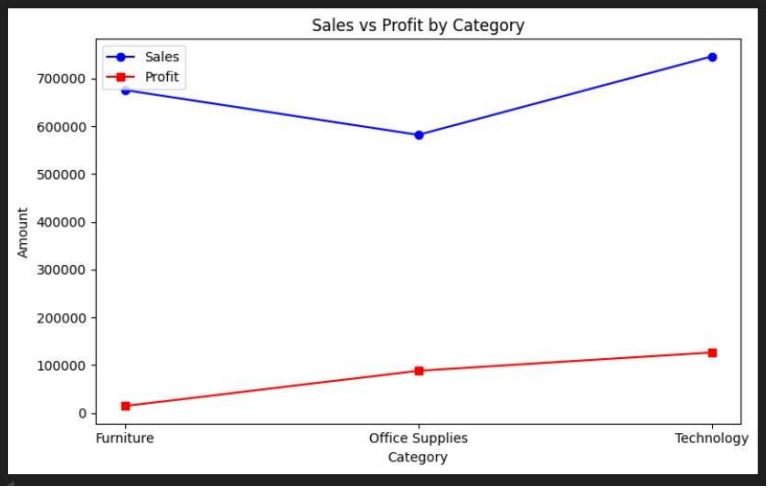
✓ 0.3s



```python
category_profit = df.groupby('Category')['Profit'].sum()

plt.figure(figsize=(8,5))
plt.plot(category_sales.index, category_sales.values, marker='o', label='Sales', color='blue')
plt.plot(category_profit.index, category_profit.values, marker='s', label='Profit', color='red')
plt.title('Sales vs Profit by Category')
plt.xlabel('Category')
plt.ylabel('Amount')
plt.legend()
plt.tight_layout()
plt.show()
```

✓ 1.0s



Sign: _____.

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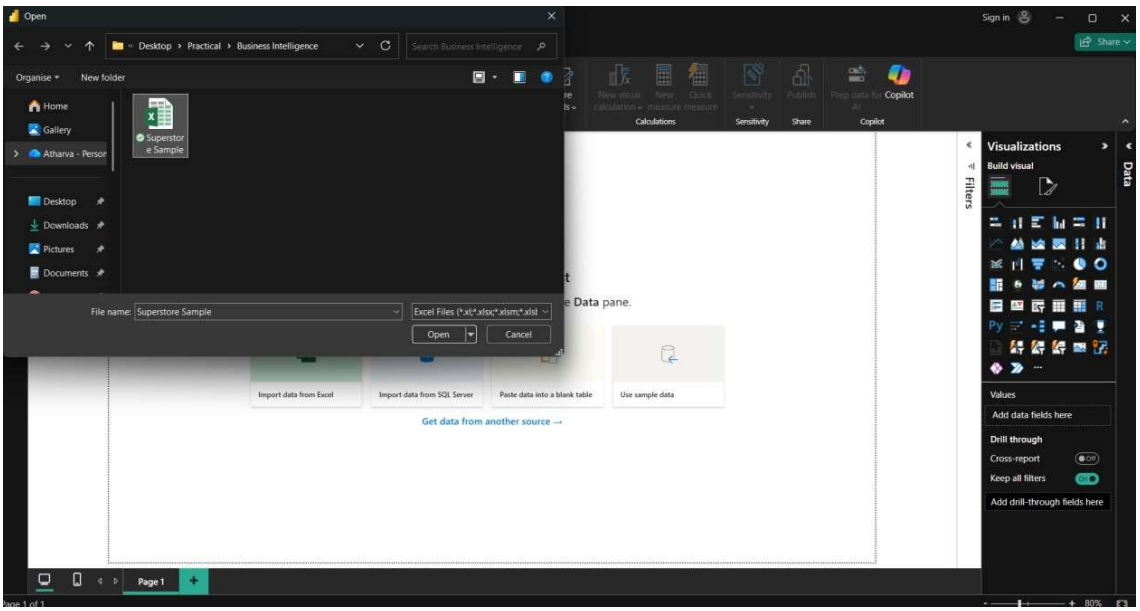
Laboratory Journal | Subject: Computer Laboratory – IV **AY:2025-26**

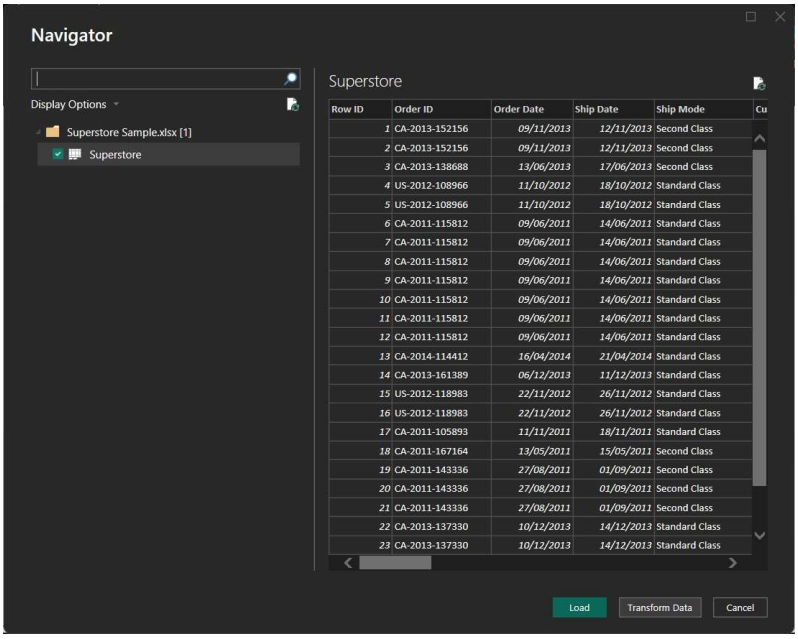
Name: Asad Alwase **Class:** B.E. (A) **Roll No:** 22542

Lab Practical 9

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

STEP 1: Extraction (Load Data)

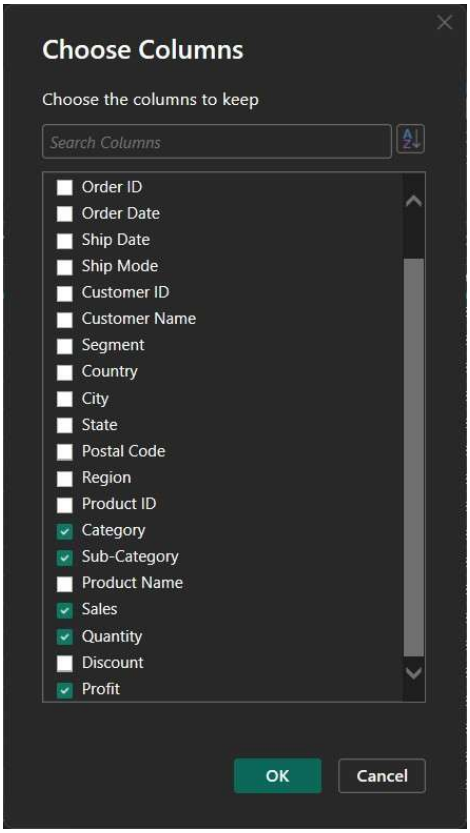




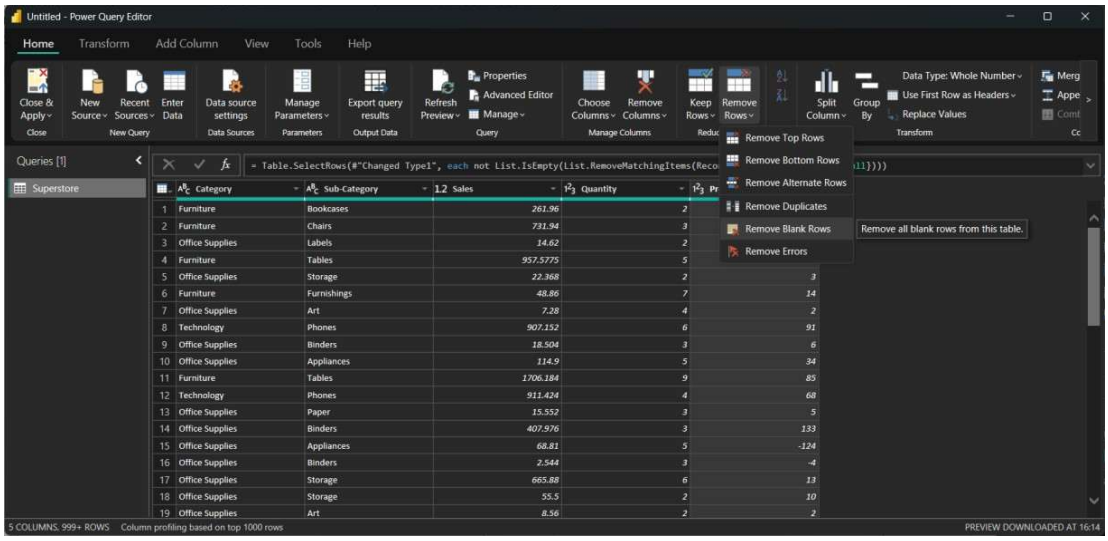
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Cu
1	CA-2013-152156	09/11/2013	12/11/2013	Second Class	
2	CA-2013-152156	09/11/2013	12/11/2013	Second Class	
3	CA-2013-138688	13/06/2013	17/06/2013	Second Class	
4	US-2012-108966	11/10/2012	18/10/2012	Standard Class	
5	US-2012-108966	11/10/2012	18/10/2012	Standard Class	
6	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
7	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
8	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
9	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
10	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
11	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
12	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
13	CA-2014-114412	16/04/2014	21/04/2014	Standard Class	
14	CA-2013-161389	06/12/2013	11/12/2013	Standard Class	
15	US-2012-118983	22/11/2012	26/11/2012	Standard Class	
16	US-2012-118983	22/11/2012	26/11/2012	Standard Class	
17	CA-2011-105893	11/11/2011	18/11/2011	Standard Class	
18	CA-2011-167164	13/05/2011	15/05/2011	Second Class	
19	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
20	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
21	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
22	CA-2013-137330	10/12/2013	14/12/2013	Standard Class	
23	CA-2013-137330	10/12/2013	14/12/2013	Standard Class	

STEP 2: Transformation

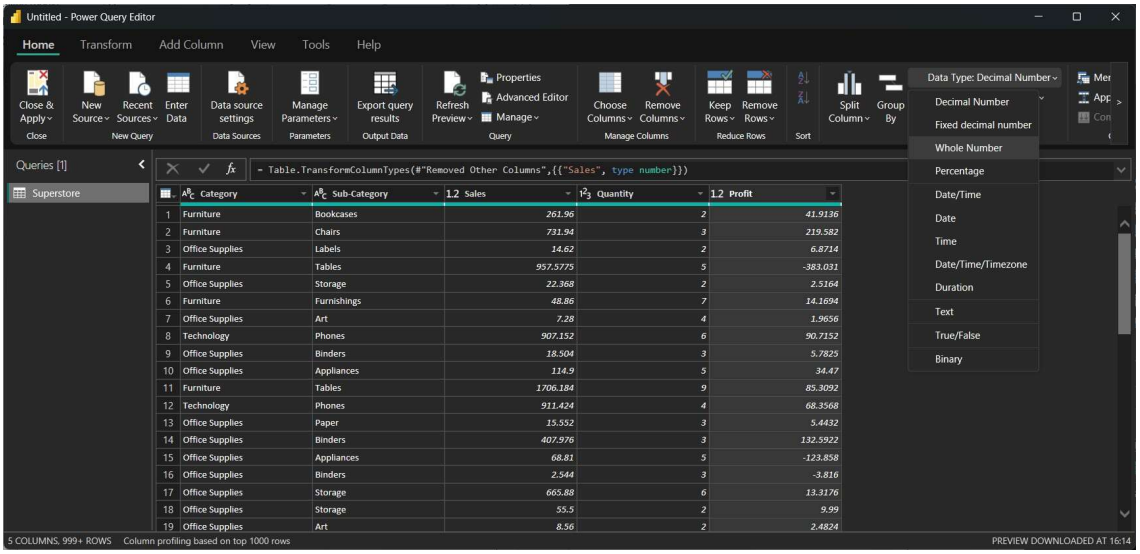
(A) Remove Unnecessary Columns (Select the ones you want to keep)



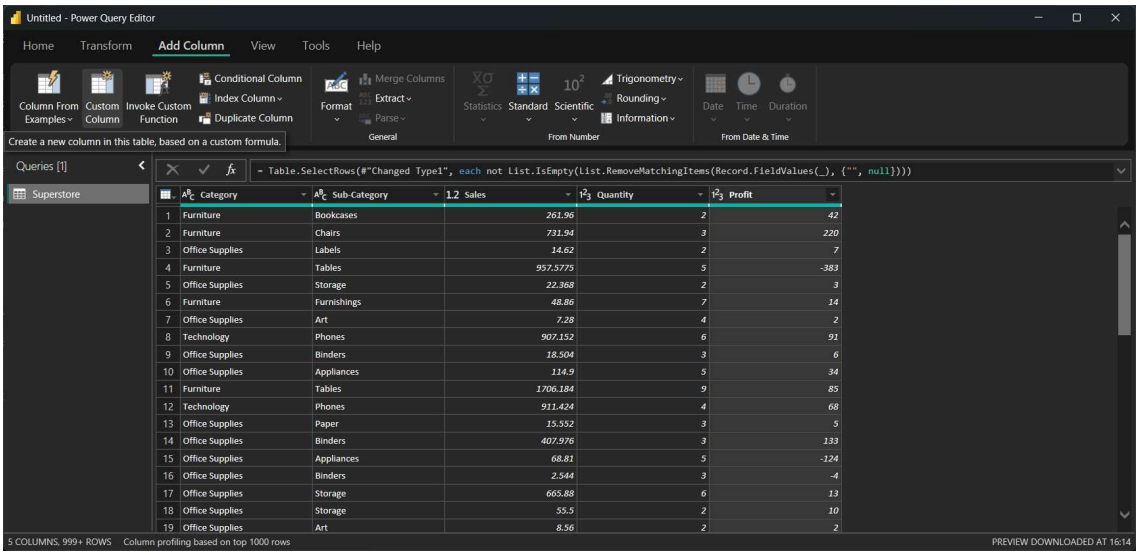
(B) Handle Missing Values

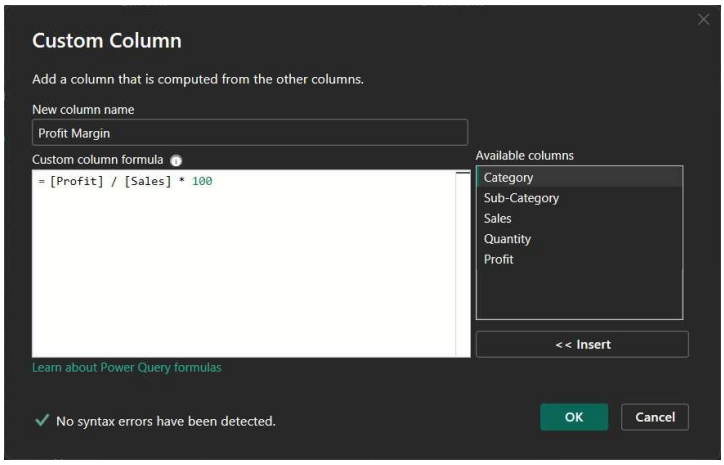


(C) Changing the Data Type:

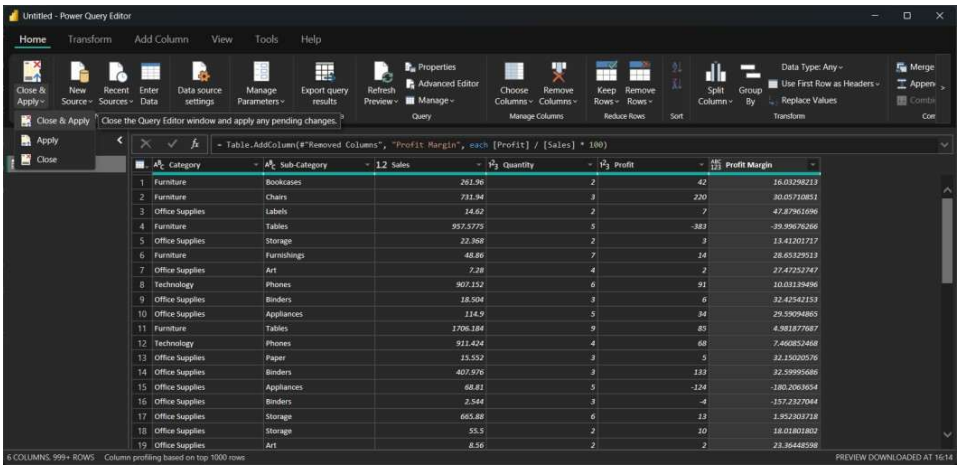


(D) Create NEW Column

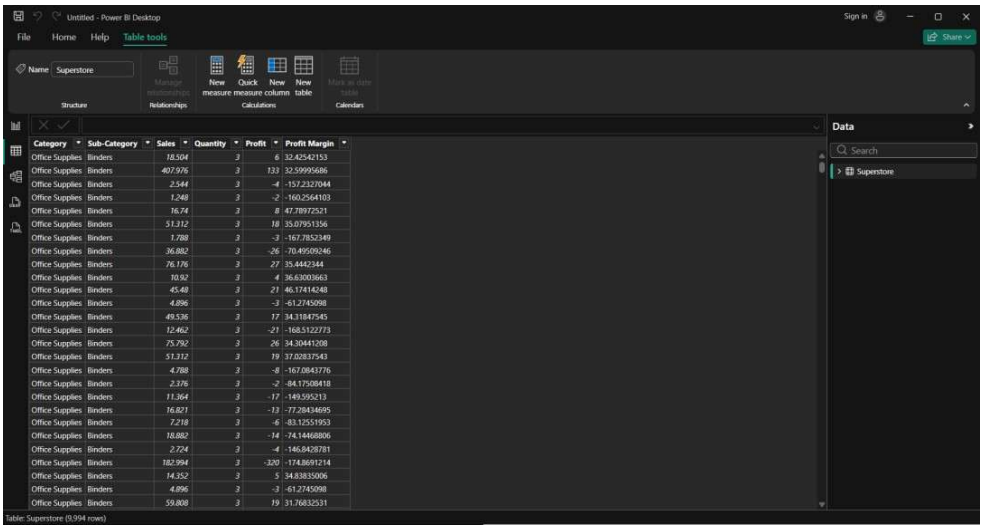


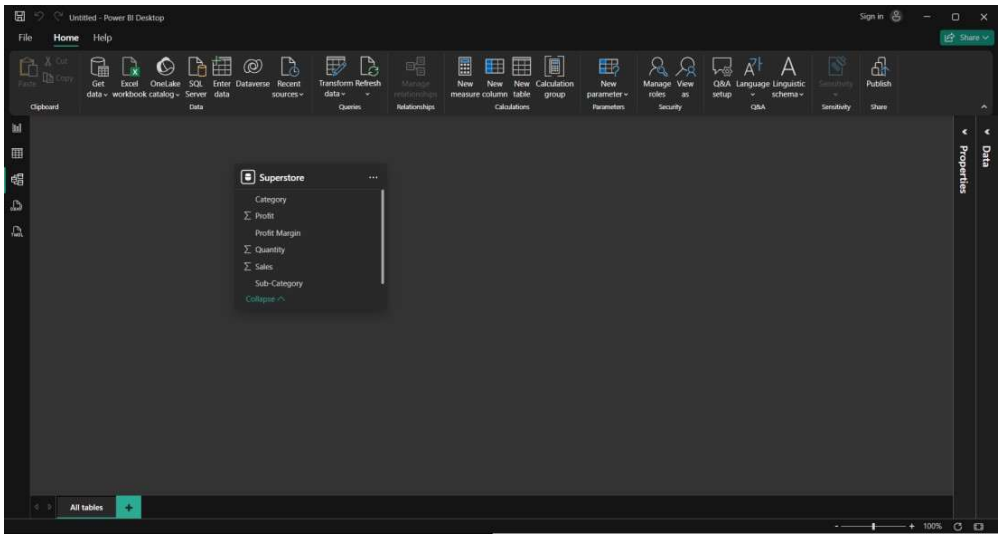


STEP 3: Loading



STEP 4: Database Construction




Untitled - Power BI Desktop
Sign in
File
Home
Help
Share
Cut
Copy
Paste
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
Manage relationships
Relationships
New measure
New column
New table
Calculation group
Calculations
New parameter
Parameters
Manage roles
View as
Security
Q&A setup
Language
Linguistic schema
Q&A
Sensitivity
Publish
Share
Superstore
Category
Profit
Profit Margin
Quantity
Sales
Sub-Category
Collapse
Properties
Data
All tables
100%

Sign: _____.